

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 1 – INDUSTRY PROBLEM TASK (5 Questions)**

|                         |                                                                                                                         |                      |                                           |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------|
| <b>Course Code:</b>     | DSA06                                                                                                                   | <b>Course Title:</b> | Data Handling and Visualization           |
| <b>CO Assessed:</b>     | CO2 – Precision (BL3)                                                                                                   | <b>Assessment:</b>   | Assessment Tool 1 – Industry Problem Task |
| <b>Weightage:</b>       | 40% of CIA                                                                                                              | <b>Total Marks:</b>  | 50 (5 Questions × 10 Marks)               |
| <b>CO2 Statement:</b>   | <i>Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)</i> |                      |                                           |
| <b>Student Name:</b>    | RENEE VINCENT ASR                                                                                                       | <b>Reg. No.:</b>     | 192424161                                 |
| <b>Section / Batch:</b> | BTECH AI&DS                                                                                                             | <b>Date:</b>         |                                           |

**Code of Conduct**

*I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Renee Vincent

**Instructions**

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

| Section / Topic                                | Focus                                                                                                    | Q Nos           | Marks |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------|-------|
| <b>Categorical Data Visualization</b>          | Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts | 1               | 10    |
| <b>Distribution Analysis</b>                   | Understanding defect distribution and variability using histograms, boxplots, and violin plots           | 2               | 10    |
| <b>Relationship Visualization</b>              | Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques     | 3               | 10    |
| <b>Geospatial &amp; Temporal Visualization</b> | Analyzing delivery performance using maps, time-series plots, and heatmaps                               | 4               | 10    |
| <b>Distribution Comparison</b>                 | Comparing patient data distributions using ridgeline plots, density plots, and boxplots                  | 5               | 10    |
| <b>GRAND TOTAL:</b>                            |                                                                                                          | <b>50 Marks</b> |       |

## **Annexure A – Industry Problem Task**

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**1.A university wants to analyze the academic performance of students across different departments (Computer Science, Mechanical, Civil, Electronics, etc.). The administration needs to compare average scores, identify top-performing departments, and examine variations in student results.**

**Task: Design suitable visualizations to represent student scores and performance distributions. Justify the use of bar plots, grouped bars, histograms, and boxplots. Explain how these visualizations help identify performance trends and support academic improvement strategies.**

### **Aim**

To analyze the academic performance of students across different departments using appropriate data visualization techniques in R.

### **Bar Plot**

- A bar plot compares the average scores of different departments.
- Each bar represents one department.
- It helps identify the highest and lowest performing departments.
- Best suited for comparing categorical data.

### **Grouped Bar Plot**

- A grouped bar plot compares multiple subjects within each department.
- Different colored bars represent different subjects.
- It helps compare departmental performance across subjects.
- Useful for identifying subject-wise strengths and weaknesses.

### **Histogram**

- A histogram shows the distribution of student scores.
- It divides marks into intervals (bins).
- Helps identify whether scores are normally distributed, skewed, or concentrated.
- Useful for understanding overall performance patterns.

### **Box Plot**

A box plot displays the spread of scores for each department.

Shows:

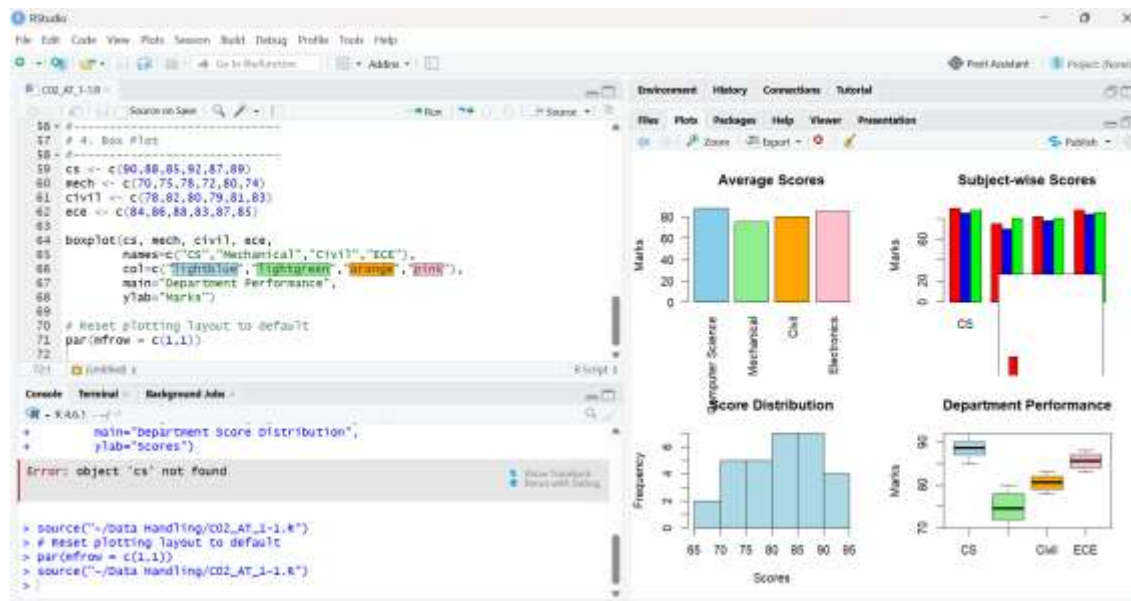
- Minimum value
- Maximum value
- Median
- First Quartile (Q1)
- Third Quartile (Q3)
- Outliers

Helps compare score variability among departments.

## Justification of Visualizations

| Visualization    | Purpose                                             |
|------------------|-----------------------------------------------------|
| Bar Plot         | Compare average scores of departments               |
| Grouped Bar Plot | Compare subject-wise performance across departments |
| Histogram        | Analyze score distribution                          |
| Box Plot         | Detect variation, median, and outliers              |

## Output



**2.A supermarket chain collects data on customer purchases across various product categories and seasons. The management wants to understand spending patterns, compare sales distributions, and identify categories with the highest contribution to revenue.**

**Task: Recommend appropriate visualization techniques such as stacked bars, heatmaps, density plots, and violin plots. Explain how these visualizations help reveal customer preferences, seasonal trends, and business opportunities.**

### Aim

To analyze customer purchase patterns across product categories and seasons using suitable data visualization techniques in R.

### Stacked Bar Plot

- A stacked bar plot compares the total sales of different product categories across seasons.
- Each bar represents a product category, while different colors represent seasons.
- It helps identify which season contributes the most to overall sales.
- Best suited for comparing the composition of sales among categories.

### Heatmap

- A heatmap represents sales data using different color intensities.
- Darker colors indicate higher sales, while lighter colors indicate lower sales.
- It helps quickly identify high-performing and low-performing category-season combinations.
- Useful for recognizing sales patterns and seasonal demand.

### Density Plot

- A density plot shows the distribution of customer spending.
- It provides a smooth curve representing spending frequency.
- Helps identify whether customers generally spend low, medium, or high amounts.
- Useful for understanding purchasing behavior and spending trends.

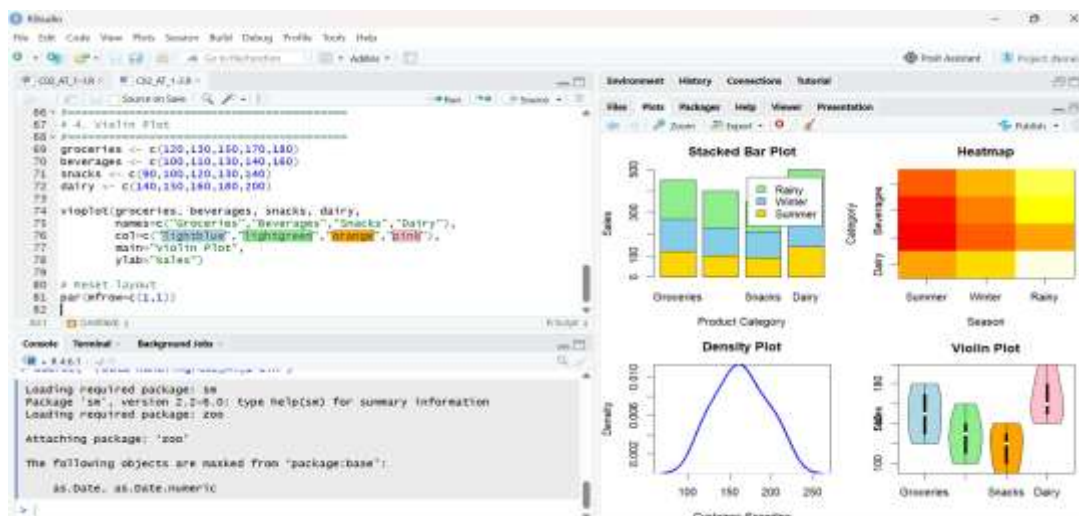
### Violin Plot

- A violin plot combines the features of a box plot and a density plot.
- It displays the distribution, spread, and concentration of sales values.
- Helps compare variability in sales across different product categories.
- Useful for detecting differences in customer purchasing patterns and identifying unusual sales distributions.

## Justification of Visualizations

| Visualization    | Purpose                                                               |
|------------------|-----------------------------------------------------------------------|
| Stacked Bar Plot | Compares seasonal sales contribution for each product category.       |
| Heatmap          | Highlights sales intensity across categories and seasons.             |
| Density Plot     | Shows the overall distribution of customer spending.                  |
| Violin Plot      | Compares the spread and variation of sales across product categories. |

## Output



**3.A smart city project gathers data on vehicle counts, traffic congestion levels, average speeds, and travel times across different locations and time periods. City planners aim to identify congestion hotspots and evaluate transportation efficiency.**

**Task: Propose visualization methods including geospatial maps, heatmaps, temporal plots, and multivariate visualizations. Describe how these visualizations can assist in traffic management, route optimization, and infrastructure planning.**

**Aim**

To analyze traffic conditions across different locations and time periods using suitable visualization techniques in R.

**Geospatial Map**

- A geospatial map displays traffic information based on geographical locations.
- It helps visualize vehicle counts, congestion levels, and traffic flow across different areas.
- Different colors or symbols represent varying traffic conditions.
- Useful for identifying congestion hotspots and planning road improvements.

**Heatmap**

- A heatmap represents traffic congestion using different color intensities.
- Darker colors indicate higher congestion, while lighter colors indicate lower congestion.
- Helps quickly identify traffic hotspots across locations and time periods.
- Useful for monitoring traffic density and peak-hour congestion.

**Temporal Plot**

- A temporal plot shows how traffic conditions change over time.
- It represents trends in vehicle counts, travel time, or average speed.
- Helps identify peak traffic hours and daily traffic patterns.
- Useful for analyzing changes in traffic flow over different time periods.

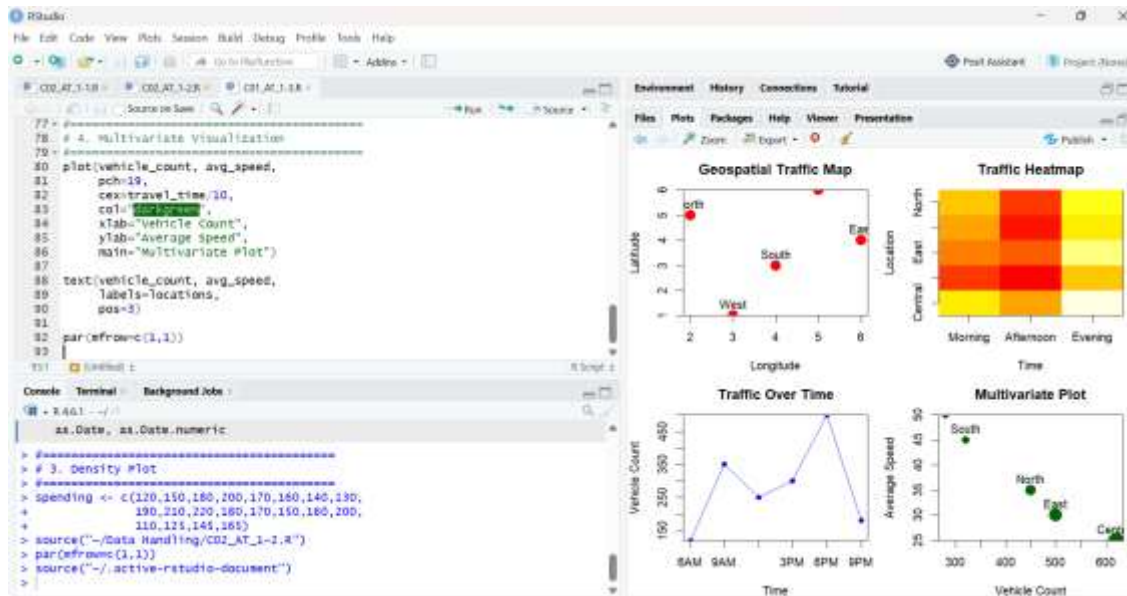
**Multivariate Visualization**

- A multivariate visualization displays multiple traffic variables together.
- It compares factors such as vehicle count, average speed, congestion level, and travel time.
- Helps understand relationships among different traffic parameters.
- Useful for evaluating transportation efficiency and making informed decisions.

**Justification of Visualizations**

| Visualization              | Purpose                                                     |
|----------------------------|-------------------------------------------------------------|
| Geospatial Map             | Displays traffic conditions across different locations.     |
| Heatmap                    | Highlights congestion intensity across locations and time.  |
| Temporal Plot              | Shows traffic trends over time.                             |
| Multivariate Visualization | Compares multiple traffic-related variables simultaneously. |

## Output



**4.A digital marketing company tracks user engagement metrics such as likes, shares, comments, viewing duration, and follower growth across multiple campaigns. The goal is to analyze relationships between engagement variables and campaign success.**

**Task: Suggest suitable visualization techniques such as scatter plots, heatmaps, network visualizations, and multivariate plots. Explain how these visualizations help identify influential factors, audience behavior patterns, and effective marketing strategies.**

### Aim

To analyze user engagement metrics across digital marketing campaigns using suitable visualization techniques in R.

### Scatter Plot

- A scatter plot displays the relationship between two engagement variables.
- Each point represents a campaign or user.
- It helps identify positive or negative correlations between metrics.
- Useful for analyzing factors that influence campaign success.

### Heatmap

- A heatmap represents engagement values using different color intensities.
- Darker colors indicate higher engagement, while lighter colors indicate lower engagement.
- Helps identify patterns across campaigns and engagement metrics.
- Useful for comparing campaign performance at a glance.

### Network Visualization

- A network visualization represents relationships among users, campaigns, or engagement metrics.
- Nodes represent entities, while edges represent connections or interactions.
- Helps identify highly connected or influential users and campaigns.
- Useful for understanding audience interactions and information flow.

### Multivariate Plot

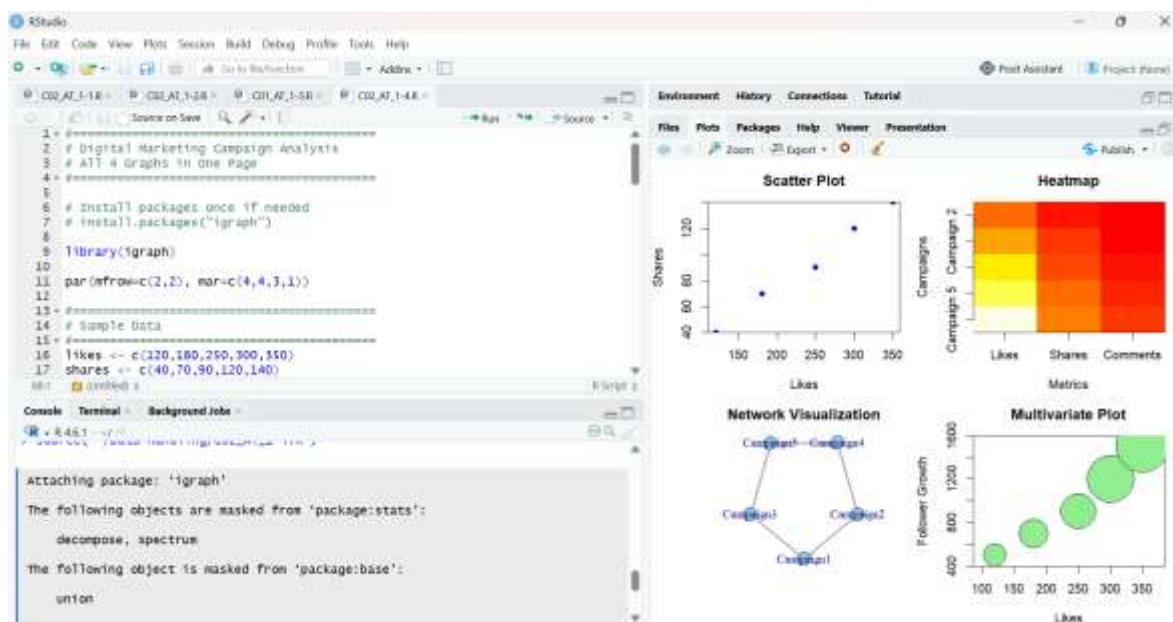
- A multivariate plot displays multiple engagement variables simultaneously.
- It compares likes, shares, comments, viewing duration, and follower growth in one visualization.
- Helps analyze relationships among several variables at the same time.

- Useful for identifying key factors contributing to campaign performance.

## Justification of Visualizations

| Visualization         | Purpose                                              |
|-----------------------|------------------------------------------------------|
| Scatter Plot          | Shows relationships between engagement variables.    |
| Heatmap               | Highlights engagement intensity across campaigns.    |
| Network Visualization | Displays connections among users or campaigns.       |
| Multivariate Plot     | Compares multiple engagement metrics simultaneously. |

## Output



**5.A healthcare organization maintains records of patients' ages, disease categories, treatment durations, recovery rates, and hospital visits. Medical administrators want to compare recovery distributions among patient groups and identify healthcare trends. Task: Select appropriate visualizations such as ridgeline plots, density plots, violin plots, histograms, and boxplots. Explain how these visualizations support disease analysis, treatment evaluation, and healthcare resource planning.**

## Aim

To analyze patient recovery patterns and healthcare trends using suitable visualization techniques in R.

## Ridgeline Plot

- A ridgeline plot displays the distribution of recovery rates across different disease categories.
- It combines multiple density plots for easy comparison.
- Helps identify similarities and differences among patient groups.
- Useful for comparing recovery patterns across diseases.

## Density Plot



- A density plot shows the distribution of recovery rates or treatment durations.
- It provides a smooth representation of the data distribution.
- Helps identify common recovery trends and variations.
- Useful for understanding patient outcome patterns.

### Violin Plot

- A violin plot combines a box plot with a density plot.
- It displays the spread, distribution, and concentration of recovery rates.
- Helps compare recovery patterns among different disease categories.
- Useful for detecting variability and unusual observations.

### Histogram

- A histogram shows the frequency distribution of patient ages or treatment durations.
- It groups values into intervals (bins).
- Helps identify common age groups or treatment durations.
- Useful for understanding overall patient characteristics.

### Box Plot

- A box plot summarizes the distribution of recovery rates or hospital visits.
- It shows the minimum, maximum, median, quartiles, and outliers.
- Helps compare recovery performance across patient groups.
- Useful for detecting unusual cases and measuring variability.

## Justification of Visualizations

| Visualization         | Purpose                                                             |
|-----------------------|---------------------------------------------------------------------|
| <b>Ridgeline Plot</b> | Compares recovery distributions across disease categories.          |
| <b>Density Plot</b>   | Shows the distribution of recovery rates or treatment duration.     |
| <b>Violin Plot</b>    | Compares variability in recovery among patient groups.              |
| <b>Histogram</b>      | Displays the frequency distribution of ages or treatment durations. |
| <b>Box Plot</b>       | Summarizes recovery rates and identifies outliers.                  |

## Output





## Rubrics for Calculation

| Criteria                            | Weight age | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               | Marks |  |
|-------------------------------------|------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|-------|--|
| Understanding of Industry Problem   | 20%        | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |       |  |
| Application of Engineering Concepts | 25%        | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |       |  |
| Solution Design                     | 25%        | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |       |  |
| Use of Tools                        | 15%        | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |       |  |
| Analysis                            | 10%        | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |       |  |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |